



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC4120
Course title (English)	Design and Analysis of Algorithms
Course title (Chinese)	算法设计及分析
Units	3
Language of Instruction	English
Description (English)	Basics of algorithm analysis: correctness and time complexity, solving recurrences. Techniques for designing efficient algorithms: divide-and-conquer, dynamic programming. Fundamental search and graph algorithms: binary trees, hashing, graph traversals, shortest paths. Introduction to complexity theory: polynomial-time reductions and NP-completeness.
Description (Chinese)	算法分析基础：正确性和时间复杂度，求解递归。设计高效算法的技术：分而治之，动态规划。基本搜索和图算法：二叉树、散列、图遍历、最短路径。复杂性理论简介：多项式时间约简和NP完全性。

2. Prerequisites / Co-requisites

A. Prerequisites

CSC3001 Discrete Mathematics;
CSC3100 Data Structures * Co-requisite or prerequisite

B. Co-requisites

CSC3100 Data Structures * Co-requisite or prerequisite

3. Learning Outcomes

Students who complete the course will have demonstrated the following:

- (1) The knowledge of algorithm design paradigms and the language of algorithm



analysis;

(2) The ability to identify, design and implement multiple algorithms based upon design paradigms such as divide-and-conquer, or dynamic programming, and relate them to the specific needs of applications;

(3) The ability to analyze the algorithmic complexity of algorithms and employ mathematical, analytical, and asymptotic notation to describe their relative merits;

(4) The understanding of the theory of NP and knowledge of handling the NP-Hard problems.

4. Course syllabus

Introduction to algorithms: Correctness and complexity, Asymptotic notation, The document distance problem; Divide-and-Conquer, FFT, Peak finding, Master Theorem; Tree search algorithms: Binary search trees, AVL Trees; Linear time sorting, Hashing, Universal Hashing, Rabin-Karp algorithm; Greedy algorithms: MST, Huffman encoding, Set cover; Graph searches and shortest paths: DFS and BFS, Bellman-Ford, Dijkstra's algorithm, A*; Dynamic programming: Fibonacci numbers, Shortest paths, Rod cutting, Text justification, Parenthesization, Longest common subsequence, Edit distance, Activity scheduling, Knapsack problem, Balanced partition, Subsetsum problem; Network flows; Computational Complexity: Classification of problems: P, EXP, NP, The halting problem, Proving hardness of problems, The P vs NP question; Coping with NP-completeness: approximation algorithms, branch and bound, local search.

5. Assessment Scheme

Component/ method	% weight
Assignments and project	30
Mid-term exam	30
Final exam	40

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research.



	Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

Formal Course and Teaching Evaluation

Informal feedback to instructor and/or teaching assistant

Departmental retreat and programme review

8. Reading

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan



Week	Content/ topic/ activity
00000 00001	Introduction, asymptotic complexity
00000 00002	Divide-and-conquer, peak finding, vEB trees, FFT
00000 00003	Master Theorem for recurrences, heaps, linear time sorting
00000 00004	Binary search trees, AVL trees, interval trees
00000 00005	Hashing I
00000 00006	Hashing II: amortized analysis, bloom filters, perfect hashing
00000 00007	Graph searches
00000 00008	Shortest paths
00000 00009	Greedy algorithms
00000 00010	Dynamic Programming I
00000 00011	Dynamic Programming II, Network flows
00000 00012	NP completeness
00000 00013	Polynomial reductions
00000 00014	Approximation algorithms, intelligent searches.

11. Any other information

Assessment Scheme

*the number of assignments/quizzes/reports is subject to change as the instructor sees fit.

*if students get less than 35% in the weighted sum of the midterm + final exam, they will not pass the class, independently of the grades of HW and project.

12. Version date

Version number	3
As of (date)	2024/11/18